

CPro : Section C

Faculty : Charm Sharma

Lecture : 3 hrs per week, Mon, Thu 8:30 - 10:00

Lab (Practical) : 3 hrs per week, Wed 14:00 - 17:00

Tutorial : 1 hr per week

Tentative Grading Scheme

Quizzes (2)	16%
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Mid Theory	10%
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mid Lab	15%
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End Theory	15%
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End Lab	20%
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Labs	19%
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Project	5%
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Language for the course : C

Reference Books :

1. A structured approach using C
(Afyouni, Forouzan)

How does code work?

→ You write code

→ compiler translates code into machine code

→ (binary object file is created)

→ to execute, you load the object file into memory

→ executable is run

Ex. program.c (you wrote this)



compiler

(ex. gcc program.c)



program.o (or program)



memory

(this is the binary object file)

(ex. ./a.out)



(the binary file is loaded into memory)

actions of code are performed

gcc is ~~the~~^a compiler for C

→ please install if not already there

where do you write code?

many people prefer using IDEs

(Integrated Development Environment)

These are applications which often include

code editors, compilers, and debuggers.

V.S. code is a popular choice

VI, vim are common in linux

some basic bash commands (shell scripting)

ls	lists all the files in a folder
mkdir	makes a new folder
cd	changes the folder you are in
rm	delete a file
..	for parent folder
.	current folder
*	any prefix
-	previous folder
~	home folder
pwd	tells you the current folder

(directory is a more appropriate term)
(folder = directory)

Common Structure of a C file

`program.c`

```
#include <stdio.h>
```

```
int main() {
```

```
    printf("Hello world!");
```

```
    return 0;
```

```
}
```

Note

→ gives you all the std functions

→ this will be run first

→ basic print line

→ end of main

→ Indentation is not compulsory in C, but it is a good practice

→ ** → save your file before running!!


```
gcc hello.c -o customname  
./customname (customname for executable)  
gcc -Wall hello.c (lists all warnings)
```

"\n" = newline character

(\) backslash is used to escape special characters

Points to remember

1. Include Header
2. End each line with a semicolon (";")
3. Start with `int main()` and `return 0`
4. Save file each time, compile it each time
(after each change, before running?)

Errors

Compilation Error:

The problem in the code was fatal, the code will not compile.

Compilation Warning:

The problem may become severe later.

It is possible to ignore some warnings if you know why exactly that warning is happening.

Warnings should ideally be resolved. Code will still compile.

Runtime Error:

Problem arose after running the code.

Comments in C

// This is a one line comment

/* This is a multi-line comment

I can write multiple lines
in this comment */

{ } curly brackets are used for blocks of code